AI and Engineering

The Future of Engineering Shaped by AI

1. Introduction: What Is Artificial Intelligence?

- Engineers can now use AI to predict machine failures, identify defects in manufactured products, optimize mechanical components, and accelerate numerical simulations.

What exactly is artificial intelligence?

- Artificial Intelligence is a field of computer science that focuses on developing systems capable of performing tasks that normally require human intelligence.
- These tasks include learning, recognizing patterns, solving problems, understanding language, and making decisions.

What exactly is artificial intelligence?

Conventional Method

- Imagine that we have an industrial machine equipped with temperature and vibration sensors.
- In a traditional monitoring system, an engineer might write a simple instruction:
  - "If the temperature exceeds 90 degrees Celsius, activate the alarm.“
- The computer follows this instruction exactly. It does not need to learn anything. It simply applies a predefined rule.

What exactly is artificial intelligence?

AI Method

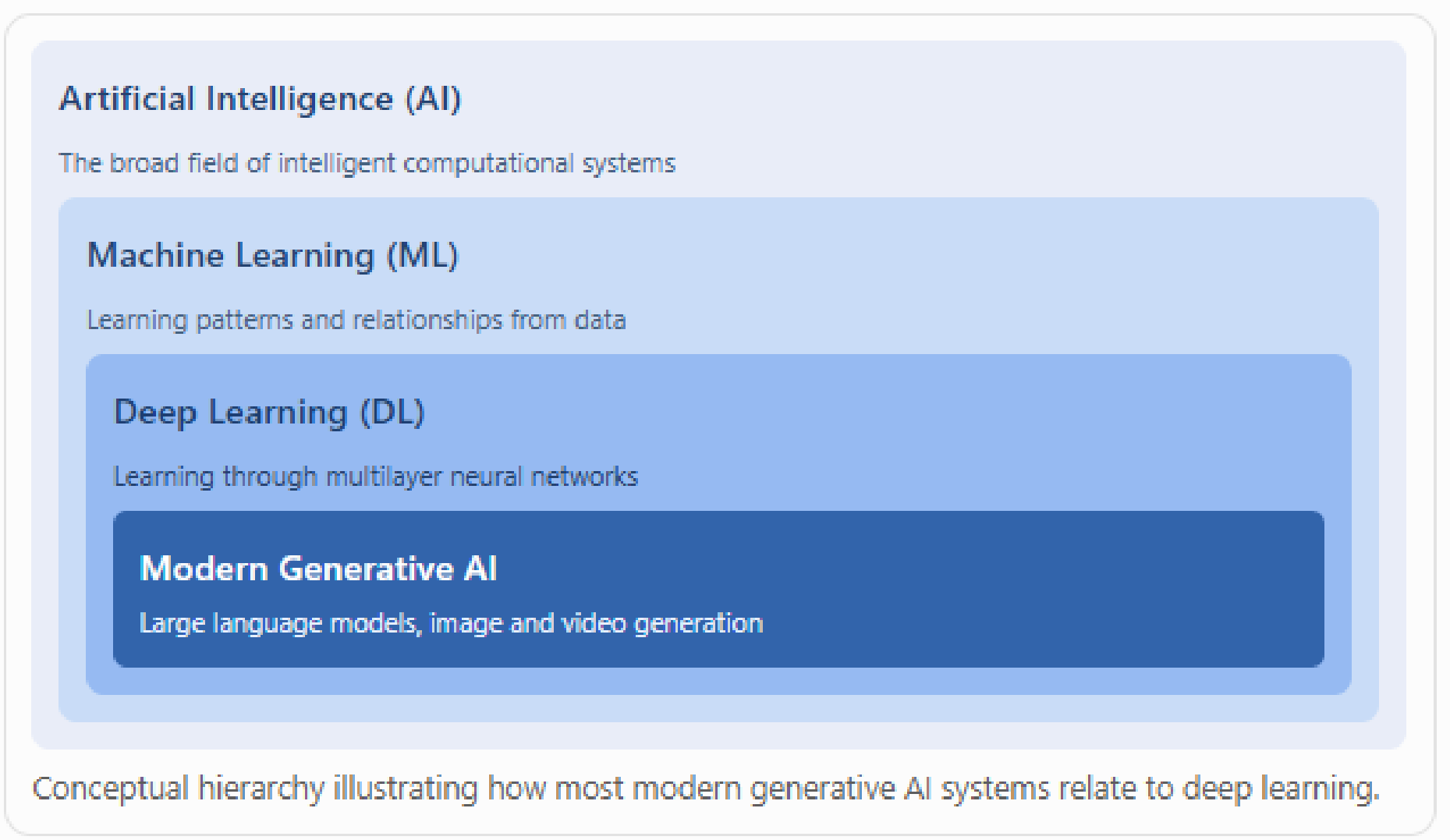
- Instead of using a fixed temperature limit, we provide the computer with thousands of measurements collected from industrial machines. Some measurements come from machines operating normally, while others come from machines that eventually failed.
- An AI model can analyze these measurements and learn relationships between temperature, vibration, operating conditions, and mechanical failures.
- After training, the model may be able to predict whether a machine is likely to fail, even before a traditional alarm is triggered.

What exactly is artificial intelligence?

- The important difference is that traditional programming follows explicitly defined rules, whereas machine learning can learn relationships from data.

- Not all artificial intelligence systems use machine learning, but machine learning is one of the most important approaches in modern AI.

Artificial Intelligence, Machine Learning, and Deep Learning



Conceptual hierarchy illustrating how most modern generative AI systems relate to deep learning.

- You have probably heard several related terms: Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI.

Artificial Intelligence, Machine Learning, and Deep Learning

- Artificial Intelligence is the broadest concept. It includes computational approaches designed to perform tasks associated with intelligent behavior.
- Machine Learning is a branch of AI in which models learn patterns from data rather than relying entirely on explicitly written rules.
- Deep Learning is a branch of machine learning that uses artificial neural networks with multiple layers to learn complex relationships.
- Finally, Generative AI refers to systems capable of producing new content, such as text, images, music, videos, and computer code.

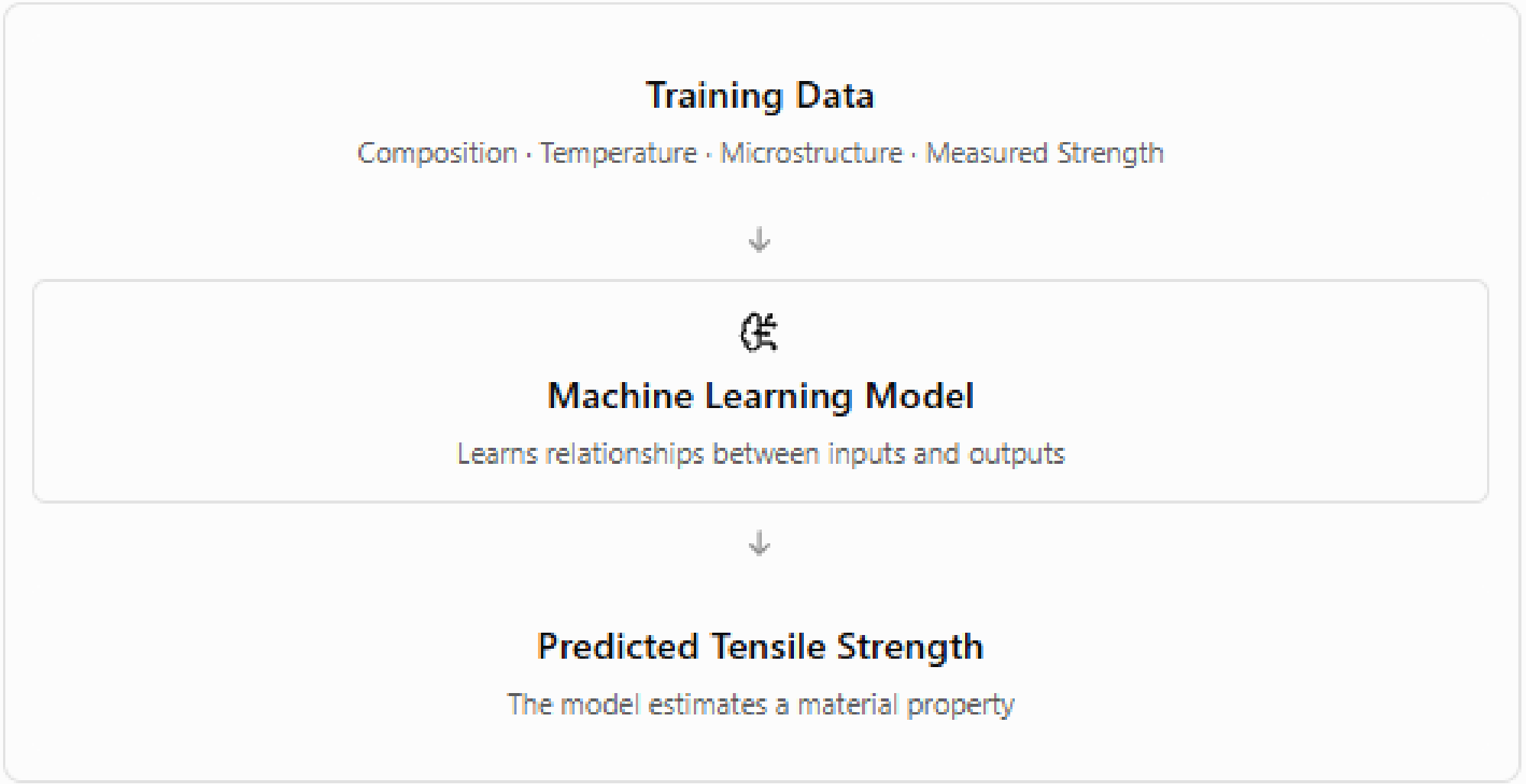
Artificial Intelligence, Machine Learning, and Deep Learning

- Most modern generative AI systems rely on deep learning.
- ChatGPT, for example, uses large language models developed using deep learning techniques.
- The important point is that AI is not a single algorithm or a single computer program. It is a broad collection of mathematical methods, computational models, and technologies.

3. How Does Machine Learning Actually Work?

- we want to develop an AI system capable of predicting the mechanical strength of a material.
- We know that the strength of a material can depend on several factors, including its chemical composition, microstructure, heat treatment, and manufacturing conditions.
- Suppose we conduct hundreds of experiments.
- For each experiment, we record the material composition, processing temperature, and measured tensile strength.
- We then use these experimental results to train a machine learning model.

3. How Does Machine Learning Actually Work?



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- At the beginning of training, the model does not know the correct relationships between these variables.
- It starts with initial parameters and produces predictions.
- For example, the actual tensile strength of a material might be 450 MPa, while the model initially predicts 380 MPa.
- The difference between the predicted and actual values produces an error.
- During training, an optimization algorithm adjusts the model parameters to reduce this error.
- This process is repeated many times using different examples.
- Gradually, the model learns relationships that help it make more accurate predictions.

3. How Does Machine Learning Actually Work?

- A common mathematical representation of a supervised learning problem is:
- $\hat{y} = f_\theta(x)$
- Here, x represents the input data, $\hat{y}$ represents the predicted output, and θ represents the parameters learned during training.
- The learning process aims to minimize a loss function, such as the mean squared error:
- $L(\theta) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$
- In simple terms, the model adjusts its parameters to reduce the difference between its predictions and the known experimental results.

3. How Does Machine Learning Actually Work?

- Once the model has been trained, it can receive new input values and predict the properties of materials that were not included in its training examples.
- However, accurate predictions depend on the quality and relevance of the training data.
- An AI model trained on steel alloys, for example, should not automatically be expected to make reliable predictions for polymer composites.

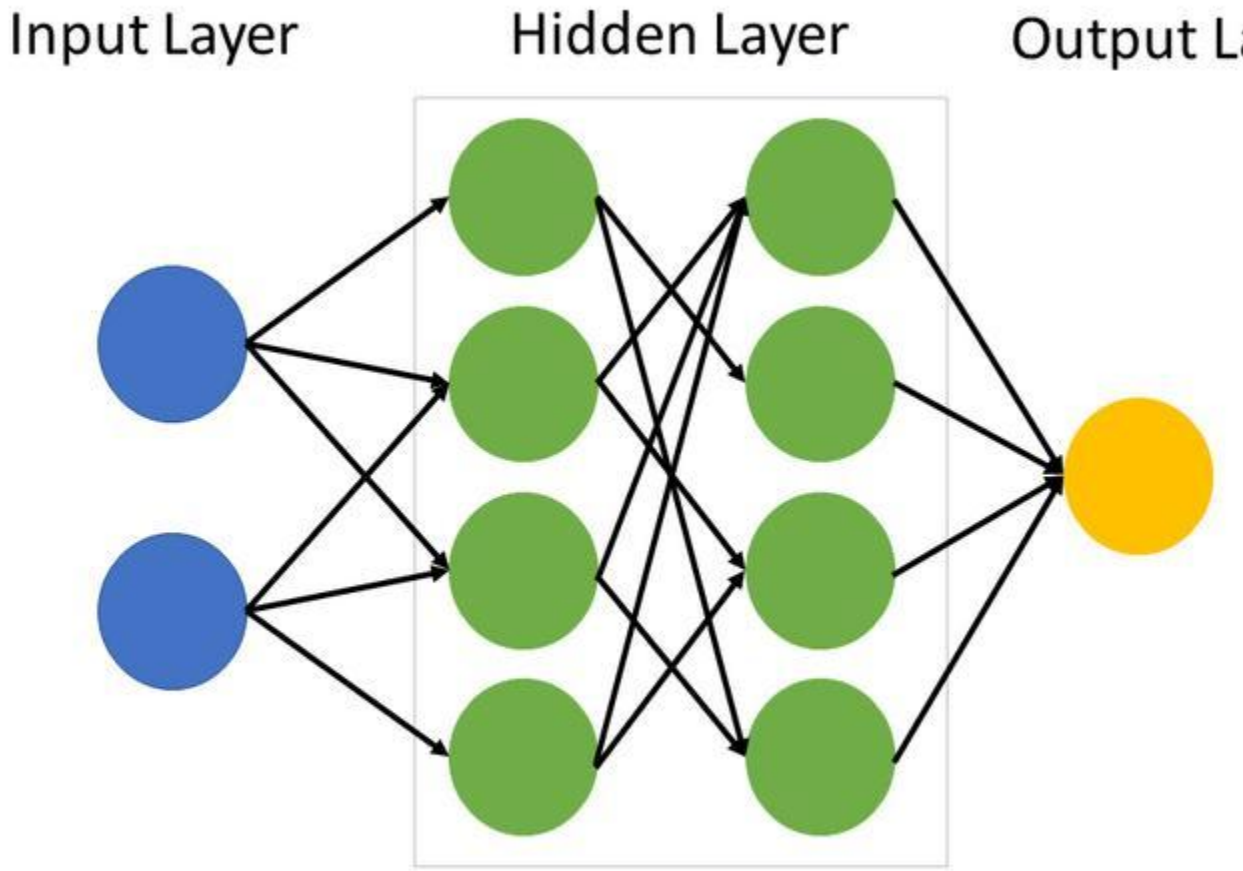
3. How Does Machine Learning Actually Work?

- **Training and Inference**
- We must distinguish between two important processes.
- During training, the model learns by adjusting its parameters using data.
- During inference, the trained model uses its learned parameters to make predictions for new inputs.
- This distinction is fundamental to understanding how modern AI systems operate.

4. How Do Artificial Neural Networks Work?

- One of the most important technologies behind modern artificial intelligence is the artificial neural network.
- Artificial neural networks are computational models loosely inspired by the structure of biological neural systems.
- However, an artificial neural network is not a digital copy of the human brain. It is a mathematical system consisting of interconnected computational units.

4. How Do Artificial Neural Networks Work?

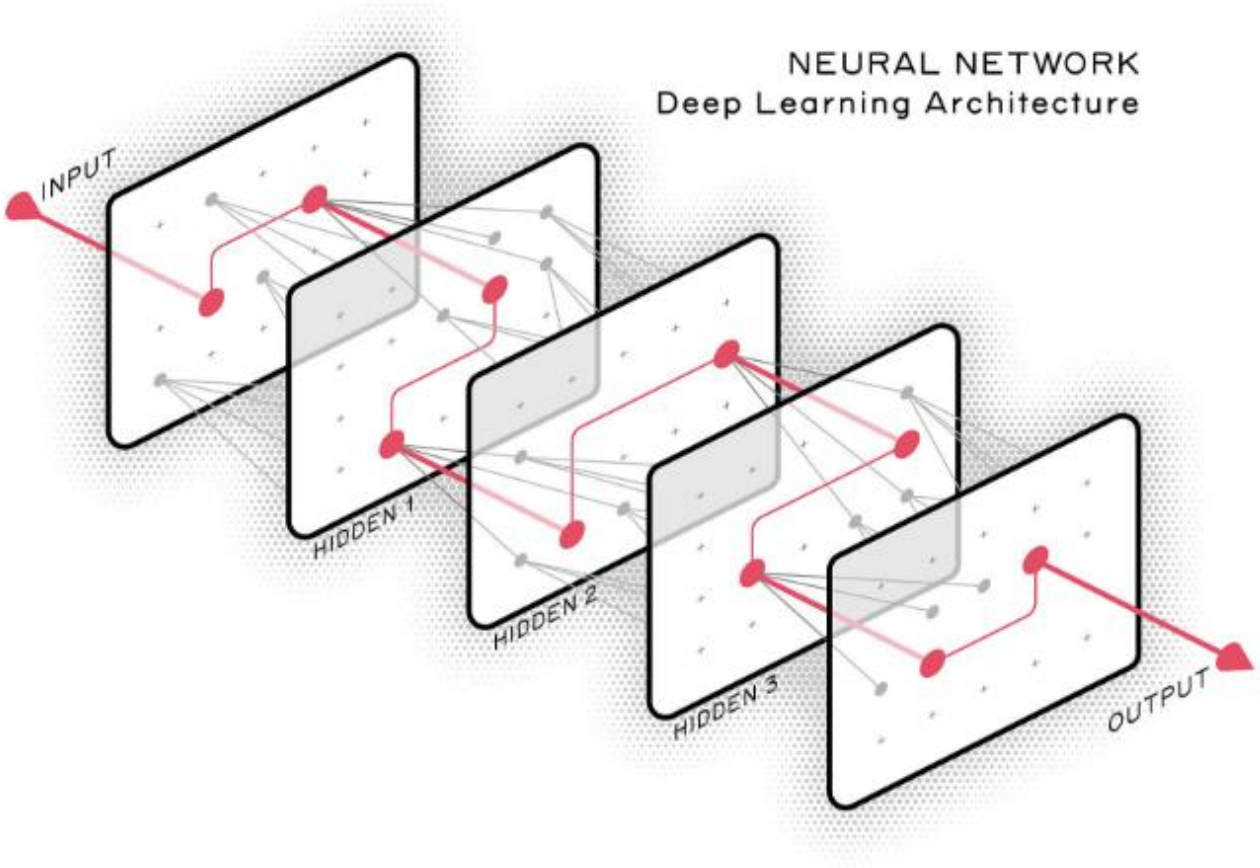


Artificial Neural Networks

4. How Do Artificial Neural Networks Work?

- A typical neural network contains three main components: an input layer, one or more hidden layers, and an output layer.
- The input layer receives the information we provide to the model.
- For example, suppose we want to estimate the tensile strength of a metal.
- The inputs might include temperature, chemical composition, grain size, and manufacturing parameters.
- These values pass through interconnected artificial neurons.

How a Hidden Layer Works Mathematically



How a Hidden Layer Works Mathematically

- In artificial neural networks (ANNs), **hidden layers** are the intermediate layers of nodes (artificial neurons) located between the **input layer** (which receives raw data) and the **output layer** (which produces the final prediction).
- They are called "hidden" because their values are not directly observed in the training dataset—you only provide the inputs and the desired outputs, while the network figures out the internal representations on its own

How a Hidden Layer Works Mathematically

- **Linear Combination:** The layer takes the vector of outputs from the previous layer ($\mathbf{x}$), multiplies it by a **weight matrix** ($\mathbf{W}$), and adds a **bias vector** ($\mathbf{b}$).
- **Non-Linear Activation:** The result is passed through a non-linear **activation function** (σ), such as ReLU, Sigmoid, or $\tanh$.
- $h = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$
- **Why non-linearity matters:** Without non-linear activation functions in the hidden layers, stacking multiple layers would be mathematically pointless. The product of multiple linear matrices is just another single linear matrix ($\mathbf{W}_2(\mathbf{W}_1\mathbf{x}) = \mathbf{W}_{\text{eff}}\mathbf{x}$), meaning the entire network would collapse into a simple linear regression model incapable of learning complex curves or physical behaviors.

How a Hidden Layer Works Mathematically

- **Core Functions of Hidden Layers**
- **Hierarchical Feature Extraction:** Each successive hidden layer learns increasingly abstract features. In an engineering context (like predicting material failure or fluid flow), early hidden layers might detect simple local gradients or relationships between stress and strain, while deeper layers combine those into complex non-linear phenomena like plastic yielding, turbulence, or geometric stress concentrations.
- **Universal Approximation:** According to the *Universal Approximation Theorem*, a neural network with even a single hidden layer containing enough neurons can approximate any continuous function to arbitrary accuracy. Adding more hidden layers (making the network **"deep"**) allows it to represent highly complex functions much more efficiently with fewer total parameters.
- **Dimensionality Transformation:** Hidden layers project input data into higher- or lower-dimensional spaces where data points that were previously entangled become linearly separable or easier to map to a target output.

How a Hidden Layer Works Mathematically



4. How Do Artificial Neural Networks Work?

- A simplified artificial neuron can be represented as:
- $y = f(\sum_{i=1}^{n} w_i \, x_i + b)$
- Here:
- x_i represents the input values.
- $w_i$represents the weights associated with those inputs.
- b is a bias parameter.
- f is an activation function.
- y is the neuron's output.

4. How Do Artificial Neural Networks Work?

- What does this mean?
- Each input contributes to the final result according to its associated weight.
- During training, the neural network adjusts its weights and biases to improve its predictions.
- When many neurons are connected across multiple layers, the network can learn complicated nonlinear relationships.
- This is particularly useful in engineering.
- For instance, the relationship between a material's microstructure and its mechanical properties may be extremely complex.
- A neural network can learn statistical relationships between microstructural features and measured mechanical performance.
- Such models can support materials research, manufacturing optimization, and engineering analysis.
- The important concept is that a neural network learns primarily by adjusting numerical parameters, not by memorizing a list of answers.

Does Artificial Intelligence Actually Understand Engineering?

- If an AI system can solve an engineering problem correctly, does that mean it understands engineering in the same way a human engineer does?
- Let us examine a simple problem.
- Suppose a steel rod has a cross-sectional area of 100 mm^2 and is subjected to a tensile force of 10 kN.
- We want to calculate the normal stress.

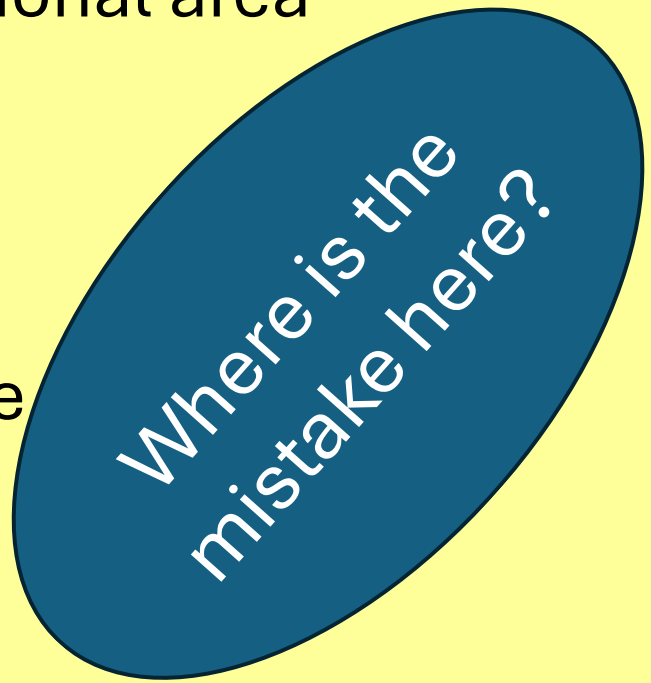
Does Artificial Intelligence Actually Understand Engineering?

- We know that normal stress is calculated as:
- $\sigma = \frac{F}{A}$
- Substituting the values:
- $\sigma = \frac{10\,000\ \mathrm{N}}{100\ \mathrm{mm}^2}$
- Therefore:
- $\sigma = 100\ \mathrm{MPa}$
- ChatGPT can often solve this problem correctly.



Does Artificial Intelligence Actually Understand Engineering?

- But imagine that instead of 100 mm^2, the cross-sectional area were given as 100 cm^2.
- Would the result be the same?
- Of course not.
- The new cross-sectional area is 10,000 mm^2, and the corresponding stress would be:
- $\sigma = \frac{10\,000\ \mathrm{N}}{10\,000\ \mathrm{mm^2}} = 1\ \mathrm{MPa}$
- A small change in units produces a major difference in the result.



Does Artificial Intelligence Actually Understand Engineering?

- This example illustrates why engineering knowledge remains essential.
- AI models can produce correct answers, but they can also make mistakes involving units, assumptions, equations, and physical interpretations.
- A convincing explanation is not necessarily a correct explanation.
- An engineer must therefore be able to evaluate whether an AI-generated result is mathematically consistent and physically meaningful.

Can AI Learn Everything?

- Another important question is whether artificial intelligence can learn any problem as long as we provide enough data.
- The answer is more complicated.
- Machine learning systems depend on several factors, including the quality of the data, the suitability of the model, and the relationship between the training data and the problem being solved.
- Consider an AI model trained to predict the mechanical behavior of a particular aluminum alloy at room temperature.
- What happens if we ask the same model to predict its behavior at 800°C?
- If the model has never encountered comparable conditions during training, its prediction may be unreliable.
- This is especially important in engineering, where incorrect predictions can lead to structural failure, equipment damage, or safety problems.

Can AI Learn Everything?

- AI models are powerful, but they do not automatically know every physical law or every possible operating condition.
- For this reason, engineers increasingly investigate approaches that combine physical knowledge with machine learning.
- These approaches are often called physics-informed or physics-guided machine learning.
- The idea is to incorporate relevant physical principles, governing equations, or engineering constraints into learning and prediction processes.
- This creates an important connection between artificial intelligence and traditional engineering disciplines.

Why Is This Important for Mechanical Engineers?

- Why should mechanical engineers learn about artificial intelligence?
- The answer is that AI is increasingly becoming part of the tools and methods engineers use to solve problems.
- Mechanical engineers traditionally use mathematics, physics, experimental measurements, computer-aided design, and numerical simulations.
- Artificial intelligence introduces additional computational capabilities.
- For example, engineers can use AI to analyze experimental measurements, predict material properties, identify manufacturing defects, estimate equipment failure probabilities, and support design optimization.

Why Is This Important for Mechanical Engineers?

- However, these applications do not eliminate the importance of fundamental engineering knowledge.
- If you want to use artificial intelligence in structural analysis, you still need to understand mechanics of materials.
- If you want to use AI in thermal engineering, you need to understand thermodynamics and heat transfer.
- If you want to develop intelligent manufacturing systems, you need knowledge of manufacturing processes, materials, and machine behavior.

Why Is This Important for Mechanical Engineers?

- AI can help engineers solve problems, but engineering knowledge is essential for defining the problems correctly and evaluating the solutions.

Why Is This Important for Mechanical Engineers?

- **1-** If an AI system can solve engineering equations, do we still need to learn mathematics?
  - Consider the difference between obtaining a numerical result and understanding its physical meaning.
- **2-** Can an AI model make accurate engineering predictions without experimental data?
  - Think about the role of physical models, available information, prior training, and validation.
- **3-**If AI becomes capable of designing and analyzing mechanical systems, what will the future role of mechanical engineers be?
  - Consider problem definition, creativity, physical understanding, system integration, and engineering responsibility.

Is Artificial Intelligence Just a Statistical Machine? Can AI Be Creative?

- **1. Is AI Simply a Statistical Machine?**
- We have discussed how artificial intelligence learns from data and makes predictions. But this raises an interesting question:
- **Is artificial intelligence essentially a sophisticated statistical machine?**

• Is artificial intelligence essentially a sophisticated statistical machine?

- In many ways, yes. Much of modern artificial intelligence is built on statistics, probability, mathematics, and computational optimization.
- For example, when you ask ChatGPT a question, the underlying language model uses patterns learned during training to calculate probabilities for possible next tokens.
- It generates responses based on these learned relationships.
- However, describing AI as just a statistical machine is an oversimplification.
- Modern AI systems can combine statistical learning with logical operations, mathematical reasoning, optimization algorithms, search methods, and external computational tools.
- They can solve problems that go far beyond simple statistical predictions.
- This leads us to a much more interesting question.
- If AI learns from existing information, can it produce something genuinely new?

Can AI Be Creative?

- Can AI Be Creative?

Can AI Be Creative?

- Consider how engineers develop new ideas.
- An engineer rarely begins with absolutely no prior knowledge.
- Instead, engineers learn from existing technologies, scientific principles, previous designs, experiments, and their own experiences.
- They combine this knowledge to develop new solutions.
- For example, when designing an aircraft wing, engineers use their understanding of fluid mechanics, materials science, structural mechanics, and previous aircraft designs.

Can AI Be Creative?

- Innovation often involves combining existing knowledge in unexpected ways.
- Artificial intelligence can perform certain aspects of a similar process.
- An AI system can learn relationships from existing examples and explore new combinations of design parameters, geometries, and computational strategies.
- These combinations may lead to solutions that were not explicitly present in its training data.
- But there is an important distinction.
- **Producing a creative result does not necessarily mean experiencing inspiration in the human sense.**
- We can evaluate whether an AI-generated design is novel or useful. However, we cannot simply assume that the system experiences imagination, curiosity, or inspiration as humans do.

Can AI Invent Something That Humans Have Never Invented?

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- In 2022, researchers introduced AlphaTensor, an AI system that discovered new algorithms for matrix multiplication.
- Some of these algorithms required fewer multiplication operations than previously known methods for particular matrix sizes and arithmetic settings.
- In 2023, another AI system called AlphaDev discovered faster algorithms for sorting small sequences of data.
- These results were published in Nature.
  - **Faster sorting algorithms discovered using deep reinforcement learning**
  - **Discovering faster matrix multiplication algorithms with reinforcement learning**.

Can AI Invent Something That Humans Have Never Invented?

- Why is this important?
- Because these systems did not simply reproduce existing solutions.
- They explored large spaces of possible algorithms and discovered solutions that improved upon previously known approaches.
- This demonstrates that AI can contribute to genuine technical innovation.
- However, these discoveries were made within computational environments designed by researchers, with defined objectives and ways to evaluate solutions.

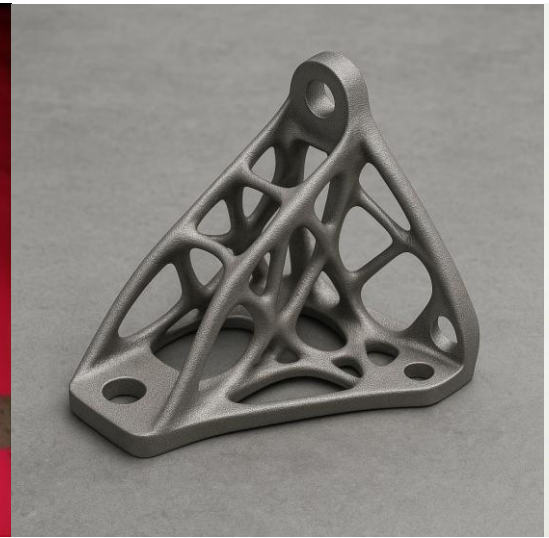
What About Mechanical Engineering?

- Now let us consider a mechanical engineering problem.
- Imagine that we want to design a lightweight structural component for an aircraft.
- Our objectives are straightforward:
  - Minimize the component's mass.
  - Maintain sufficient mechanical strength.
  - Limit deformation under operating loads.
  - Ensure that the component can be manufactured.

What About Mechanical Engineering?

- A traditional engineer might begin with familiar geometries, such as plates, beams, or reinforced brackets.
- An AI-assisted design system could instead explore thousands of possible geometries.
- Some of these geometries might look completely unfamiliar.







What About Mechanical Engineering?

- The system might suggest an unusual structure with curved supports, hollow regions, or complex internal geometries.
- Perhaps no engineer would have proposed exactly that configuration.
- If the design satisfies the engineering requirements and offers a measurable improvement, it may represent a useful innovation.
- But we must distinguish between different computational approaches. Traditional topology optimization can also generate novel geometries without using machine learning.
- The innovative result does not, by itself, prove that the algorithm uses AI.
- And whether the geometry is genuinely better must be established through appropriate engineering analysis and testing.

Is There a Difference Between Creativity and Innovation?

Concept	Meaning
Novelty	Producing something new or different.
Creativity	Producing ideas that are both original and meaningful or useful.
Innovation	Developing and applying new ideas that create practical value.
Inspiration	A subjective experience often associated with the emergence of ideas.

At the End

- If machines can also contribute to invention and discovery, how will the role of the human engineer change?